1D Transdimensional Bayesian Inversion of MT Data of the Southern African Lithosphere

A DISSERTATION

Submitted in partial fulfilment of the requirements for the award of the degree of

MASTER OF TECHNOLOGY

in

GEOPHYSICAL TECHNOLOGY

By

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12 May 2026

Candidate's Declaration

I hereby declare that the work presented in this dissertation entitled "**1D Trans-dimensional Bayesian Inversion of MT Data of the Southern African Lithosphere**" submitted in the **DEPARTMENT OF EARTH SCIENCES**, **Indian Institute of Technology Roorkee**, Roorkee, India, in partial fulfilment of the requirements for the award of the degree of **Master of Technology** in **GEOPHYSICAL TECHNOLOGY**, is an authentic record of my own work carried out under the supervision of **Prof. Arun Singh**. I have not submitted this work elsewhere for any other degree or diploma.

Date: 12 May 2026 **SHAILESH PARVADIYA**

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Prof. Arun Singh
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Abstract

Write your abstract here (250–350 words). Summarise the motivation, objectives, methodology, key results, and conclusions. Keep it self-contained with no undefined abbreviations or citations.

Keywords: keyword1, keyword2, keyword3, keyword4, keyword5

Acknowledgement

I am deeply grateful to my supervisor, **Prof. Arun Singh**, for their invaluable guidance, constant encouragement, and patient support throughout this research work.

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SHAILESH PARVADIYA
Roorkee, 12 May 2026

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List of Abbreviations

Abbreviation	**Full Form**
ABBR1	Full form of abbreviation one
ABBR2	Full form of abbreviation two

Chapter 1

Introduction

Write your opening paragraph. Introduce the broad problem, why it matters, and what gap your work fills (1).

1.1 Background and Motivation

Describe the context and motivation for your research (2).

1.2 Key Terminologies and Concepts

1.2.1 Concept / Dataset One

Define and describe the first key concept or dataset used in this thesis.

1.2.2 Concept / Dataset Two

Define and describe the second concept. Include an equation if needed:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{1.1}$$

where y_i are observed values and $\hat{y}_i$ are predicted values.

1.2.3 Concept / Dataset Three

Continue for as many concepts as needed.

1.3 Objectives of the Study

The specific objectives of this thesis are:

1. To *[state objective one]*.
2. To *[state objective two]*.
3. To *[state objective three]*.
4. To *[state objective four]*.

1.4 Organisation of the Thesis

Chapter 2 reviews relevant literature and identifies research gaps. **Chapter 3** presents the study area, datasets, and methodological framework. **Chapter 4** presents and discusses the results. **Chapter 5** summarises the conclusions and outlines future work.

Chapter 2

Literature Review

Introduce the chapter and the structure of the review.

2.1 Research Theme One

Review studies related to the first theme (1).

2.1.1 Sub-topic 1.1

Summarise relevant works and their findings.

2.1.2 Sub-topic 1.2

Continue the review.

2.2 Research Theme Two

Review studies related to the second theme.

2.2.1 Sub-topic 2.1

Summarise relevant works.

2.2.2 Sub-topic 2.2

Continue.

2.3 Research Theme Three

Review studies related to the third theme.

2.4 Research Gaps and Problem Statement

Synthesise the gaps in existing literature and state the problem this thesis addresses.

2.5 Summary

Briefly summarise what has been covered in this chapter and how it motivates the methodology.

Chapter 3

Methodology

Briefly describe the overall workflow and its components.

3.1 Study Area

Describe the geographic extent, physiography, climate, and hydrology/geology of your study area.

3.2 Data Sources

Table 3.1: Summary of datasets used in this study.

Dataset	**Variable**	**Resolution**	**Period**	**Source**
Dataset 1	Variable 1	0.25°	2002–2023	**(author?)** (1)
Dataset 2	Variable 2	1 km	2000–2023	Agency name
Dataset 3	Variable 3	Point	2004–2023	Agency name

3.2.1 Dataset One

Describe acquisition, format, and preprocessing.

3.2.2 Dataset Two

Describe acquisition, format, and preprocessing.

3.3 Method One — [Name]

Describe the first methodological component.

3.3.1 Preprocessing

Describe masking, gap handling, anomaly calculation, etc.

3.3.2 Model Architecture

Describe the model and its equations.

3.3.3 Training and Validation

Describe training strategy, hyperparameters, and validation metrics.

3.4 Method Two — [Name]

Describe the second component.

3.4.1 Feature Selection

Describe how input features were chosen.

3.4.2 Model Training and Validation

Describe the approach.

3.5 Method Three — [Name]

Describe the third component and its specific workflow.

Chapter 4

Results and Discussion

Introduce the chapter and summarise what will be presented.

4.1 Results from Method One

4.1.1 Performance Metrics

Table 4.1: Model performance metrics for Method One.

Metric	**Training**	**Validation**	**Testing**
RMSE	XX.XX	XX.XX	XX.XX
MAE	XX.XX	XX.XX	XX.XX
R^2	X.XXX	X.XXX	X.XXX

Discuss the metrics. Comment on model fit and generalisation.

4.1.2 Spatial and Temporal Patterns

Present maps and time series. Discuss seasonality and trends.

4.1.3 Comparison with Independent Data

Validate against independent observations and discuss discrepancies.

4.2 Results from Method Two

4.2.1 Model Evaluation

Report and discuss performance.

4.2.2 Spatial Maps

Present spatial outputs and interpret them.

4.2.3 Temporal Analysis

Present and discuss time series outputs.

4.3 Results from Method Three

4.3.1 Model Performance

Report performance metrics.

4.3.2 Key Findings

Discuss the physical mechanisms behind the observed patterns and implications for your application.

4.4 Synthesis and Discussion

Synthesise findings across all components. Discuss limitations and compare with existing literature (2).

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This study addressed [core problem] in [study region] using [methods]. The key conclusions are:

1. **Finding 1:** Describe the first key finding and its significance.
2. **Finding 2:** Describe the second key finding.
3. **Finding 3:** Describe the third key finding.

The developed framework provides a scalable and transferable approach for [application area].

5.2 Future Scope

- Integration of additional satellite or in-situ data sources.
- Exploration of advanced deep learning or physics-informed models.
- Systematic uncertainty quantification across the workflow.
- Extension to other regions with similar characteristics.
- Collaboration with agencies for operational implementation.

Bibliography

[1] Author, A. and Coauthor, B. (2020). *Title of a journal article.* Journal Name, 10(2), 100–115.

[2] Author, C. (2019). *Title of another paper.* Water Resources Research, 55, 1234–1256.